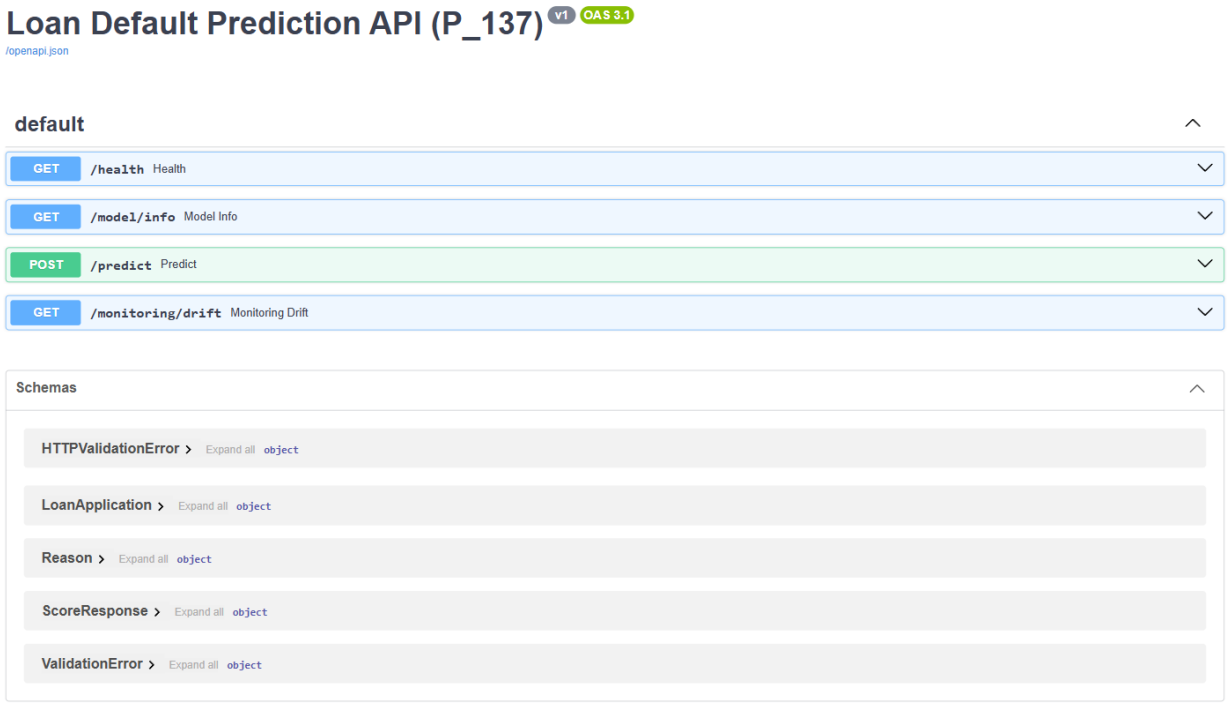
Loan Default Prediction System: Explainable Credit Risk Scoring with Drift Monitoring

HCLTech Career Shaper Programme | III Year Industry Project | Project No. 137

Output screenshots

Screens of the working system, captured on 01 November 2026 from the FastAPI service running locally at http://127.0.0.1:8000 (Swagger UI) and from the console runs of the pipeline, 1440 x 900 browser window.

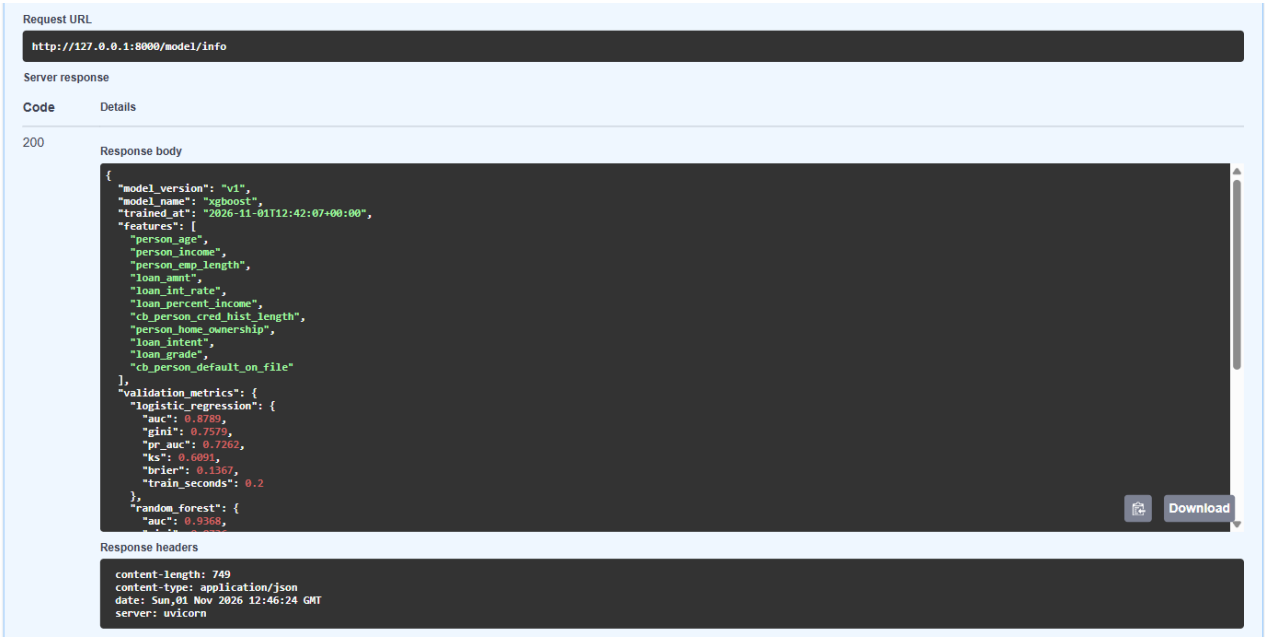
Swagger UI and service health



Screenshot 1: Swagger UI of the scoring API. FastAPI generates this interactive documentation at http://127.0.0.1:8000/docs; it lists the four endpoints (GET /health, GET /model/info, POST /predict, GET /monitoring/drift) and the request and response schemas.

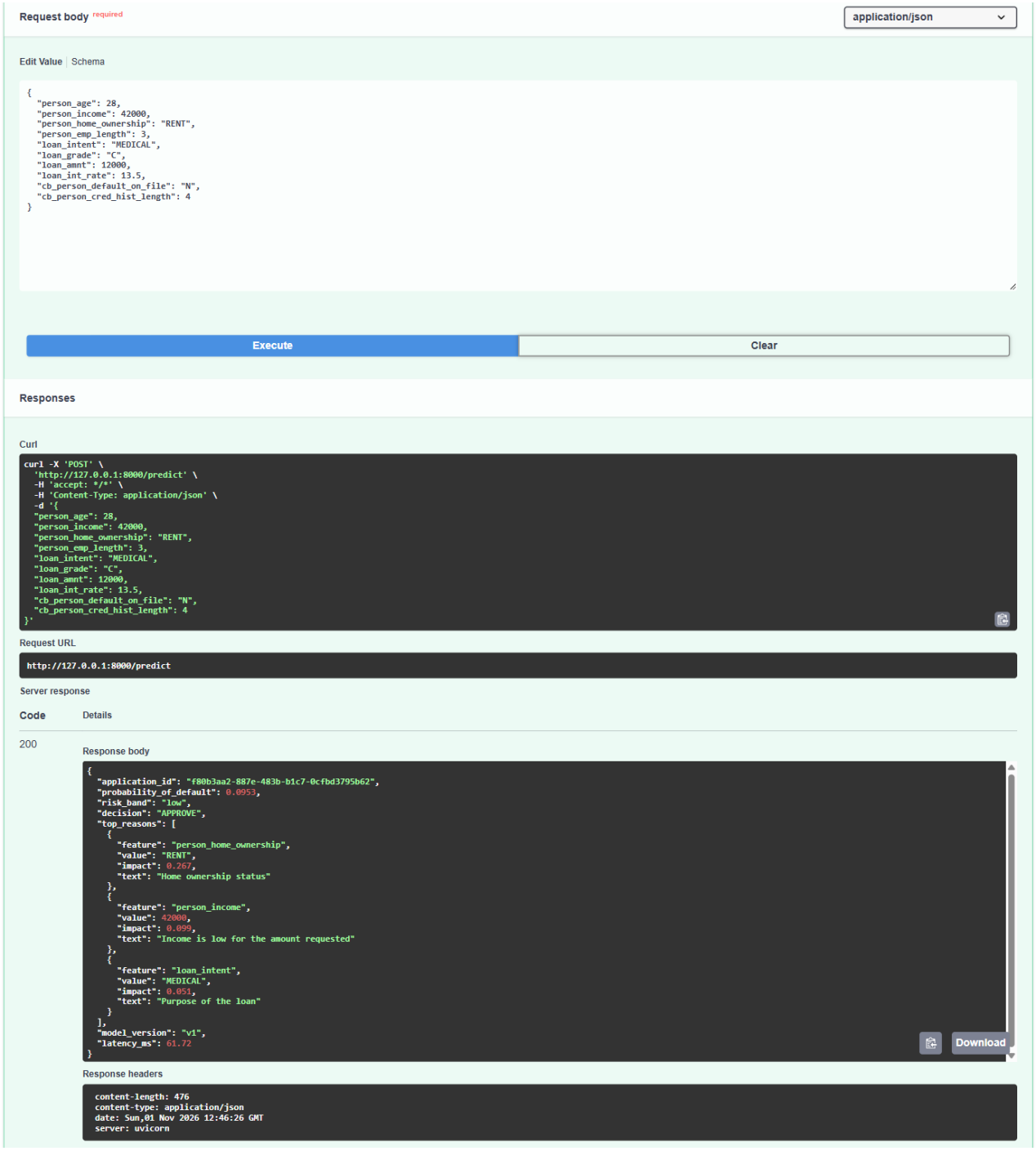


Screenshot 2: GET /health. The service answers HTTP 200 with status ok, model version v1 and kill_switch false.

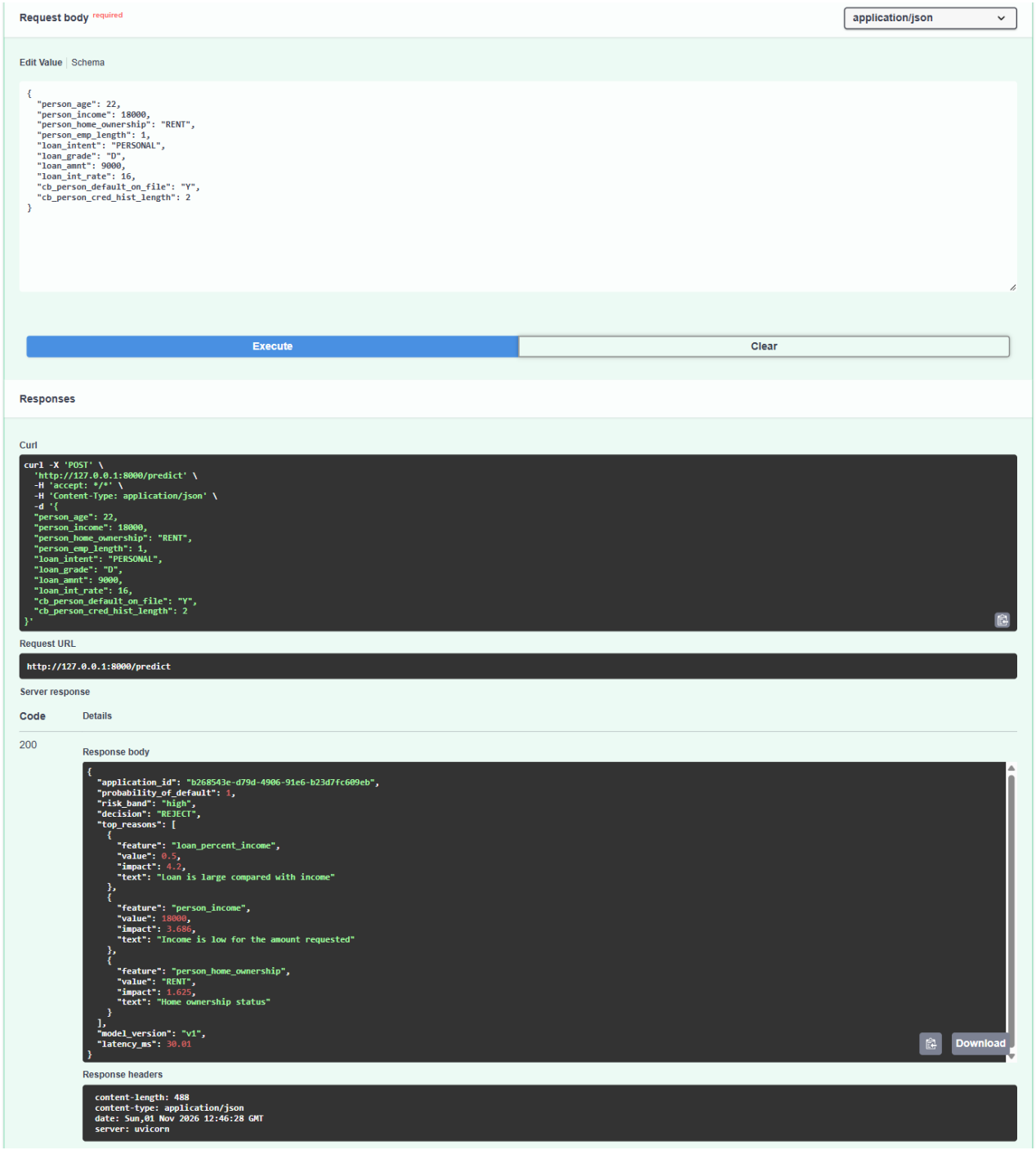


Screenshot 3: GET /model/info. Training metadata of the deployed model (xgboost, 22,686 training rows, validation AUC 0.9511 against 0.9368 for random forest and 0.8789 for logistic regression) plus the policy cut-offs approve_below 0.2 and reject_from 0.5.

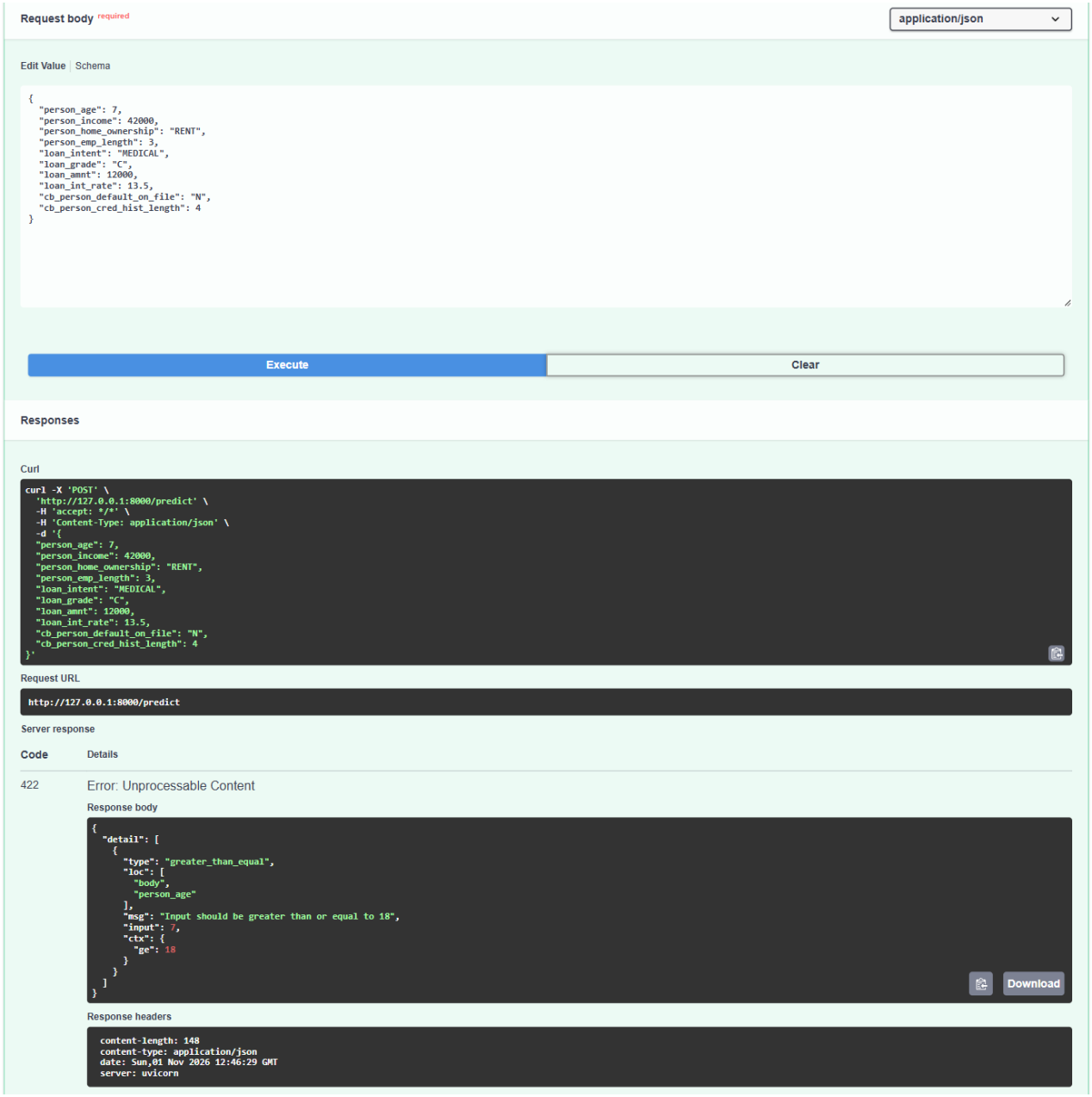
Scoring applications (POST /predict)



Screenshot 4: Low-risk applicant approved. A 28-year-old renter earning 42,000 asks for 12,000 at 13.5% (grade C): PD 0.0953, risk band low, decision APPROVE, with the three factors that still raised the risk (home ownership RENT +0.267, income +0.099, purpose MEDICAL +0.051).



Screenshot 5: High-risk applicant rejected. A 22-year-old earning 18,000 asks for 9,000 at 16% (grade D, past default Y): PD 1.0, risk band high, decision REJECT; reasons: loan is half of income (+4.2), low income (+3.686), renting (+1.625).



Screenshot 6: Invalid input rejected with HTTP 422. The same applicant with age 7 is stopped by Pydantic validation before the model runs: HTTP 422, 'Input should be greater than or equal to 18'.

Drift monitoring



Screenshot 7: Drift report refused on an empty log. With no logged predictions, GET /monitoring/drift returns HTTP 409 'Need at least 100 logged predictions, have 0', so a drift verdict is never computed on too few rows.

Request URL

```
http://127.0.0.1:8000/monitoring/drift?last_n=500
```

Server response

Code	Details
200	Response body


```
{
  "window_size": 500,
  "alerts": [],
  "decision_mix": {
    "APPROVE": 0.718,
    "REJECT": 0.184,
    "REVIEW": 0.098
  },
  "approve_baseline": 0.75,
  "features": [
    {
      "feature": "person_age",
      "psi": 0.0073,
      "status": "stable"
    },
    {
      "feature": "person_income",
      "psi": 0.019,
      "status": "stable"
    },
    {
      "feature": "person_emp_length",
      "psi": 0.0064,
      "status": "stable"
    },
    {
      "feature": "loan_amnt",
      "psi": 0.0049,
      "status": "stable"
    },
    {
      "feature": "loan_int_rate",
      "psi": 0.0122,
      "status": "stable"
    },
    {
      "feature": "loan_percent_income",
      "psi": 0.0133,
      "status": "stable"
    },
    {
      "feature": "cb_person_cred_hist_length",
      "psi": 0.0031,
      "status": "stable"
    },
    {
      "feature": "person_home_ownership",
      "psi": 0.0088,
      "status": "stable"
    },
    {
      "feature": "loan_intent",
      "psi": 0.0188,
      "status": "stable"
    },
    {
      "feature": "loan_grade",
      "psi": 0.0104,
      "status": "stable"
    },
    {
      "feature": "cb_person_default_on_file",
      "psi": 0.0001,
      "status": "stable"
    },
    {
      "feature": "model_score_PD",
      "psi": 0.0258,
      "status": "stable"
    }
  ]
}
```


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Screenshot 8: Drift check after 500 normal applications. After 500 test-set applicants were scored, every feature PSI is below 0.10 (stable), the decision mix is APPROVE 0.718 / REVIEW 0.098 / REJECT 0.184 against an approval baseline of 0.75, and the alerts list is empty.

Request URL

```
http://127.0.0.1:8000/monitoring/drift?last_n=500
```

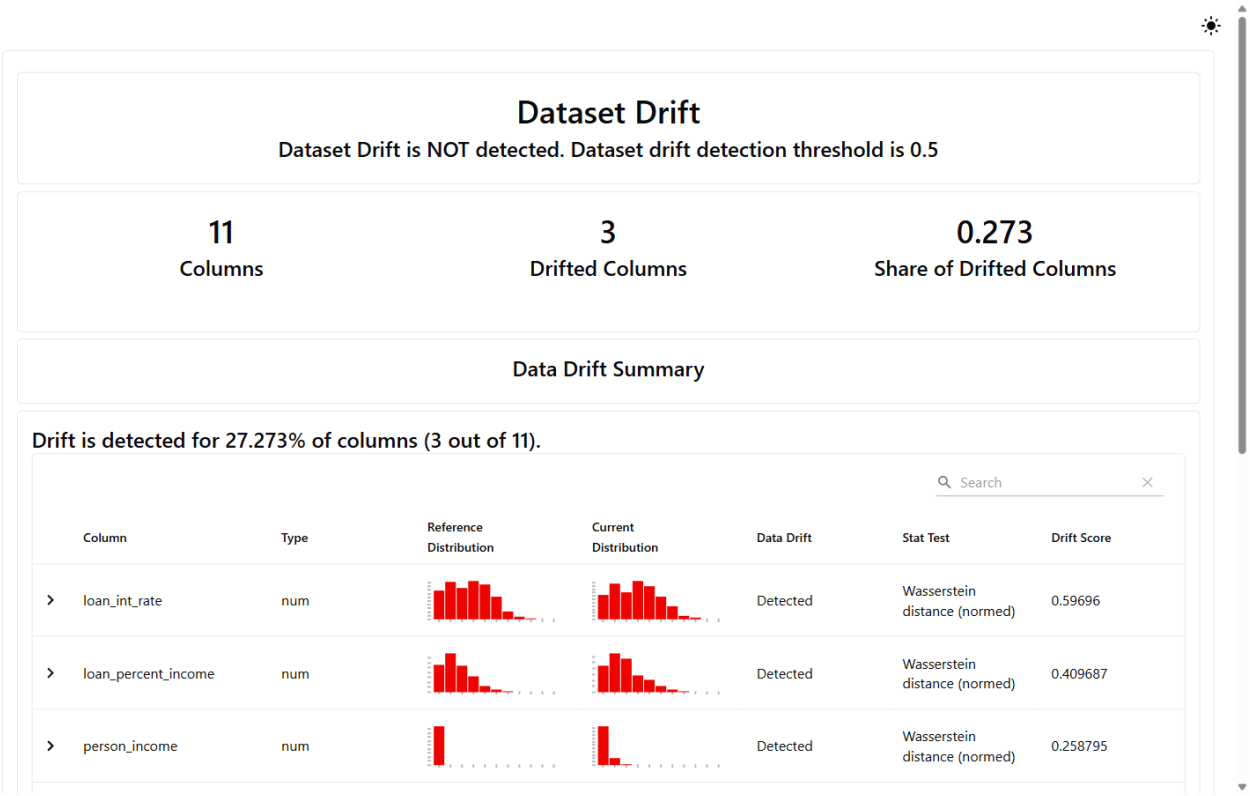
Server response

Code	Details
200	Response body


```
{
  "window_size": 500,
  "alerts": [
    "loan_int_rate",
    "decision_mix"
  ],
  "decision_mix": {
    "APPROVE": 0.572,
    "REJECT": 0.274,
    "REVIEW": 0.154
  },
  "approve_baseline": 0.75,
  "features": [
    {
      "feature": "person_age",
      "psi": 0.0073,
      "status": "stable"
    },
    {
      "feature": "person_income",
      "psi": 0.2033,
      "status": "watch"
    },
    {
      "feature": "person_emp_length",
      "psi": 0.0064,
      "status": "stable"
    },
    {
      "feature": "loan_amnt",
      "psi": 0.0049,
      "status": "stable"
    },
    {
      "feature": "loan_int_rate",
      "psi": 1.1777,
      "status": "alert"
    },
    {
      "feature": "loan_percent_income",
      "psi": 0.1234,
      "status": "watch"
    },
    {
      "feature": "cb_person_cred_hist_length",
      "psi": 0.0031,
      "status": "stable"
    },
    {
      "feature": "person_home_ownership",
      "psi": 0.0088,
      "status": "stable"
    },
    {
      "feature": "loan_intent",
      "psi": 0.0188,
      "status": "stable"
    },
    {
      "feature": "loan_grade",
      "psi": 0.0104,
      "status": "stable"
    },
    {
      "feature": "cb_person_default_on_file",
      "psi": 0.0001,
      "status": "stable"
    },
    {
      "feature": "model_score_PD",
      "psi": 0.2371,
      "status": "watch"
    }
  ]
}
```


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Screenshot 9: Drift alerts after a simulated downturn. After 500 applicants with incomes cut by 20% and rates raised by 2 points, loan_int_rate PSI is 1.1777 (alert), person_income 0.2033 and loan_percent_income 0.1234 (watch), approvals fall to 0.572, and the alerts are loan_int_rate and decision_mix.



Screenshot 10: Evidently data drift report (downturn). The Evidently 0.7 report (reports/drift_report_downturn.html) compares the stressed test set with the training data: 3 of 11 columns drifted (share 0.273), below its 0.5 dataset-level threshold.

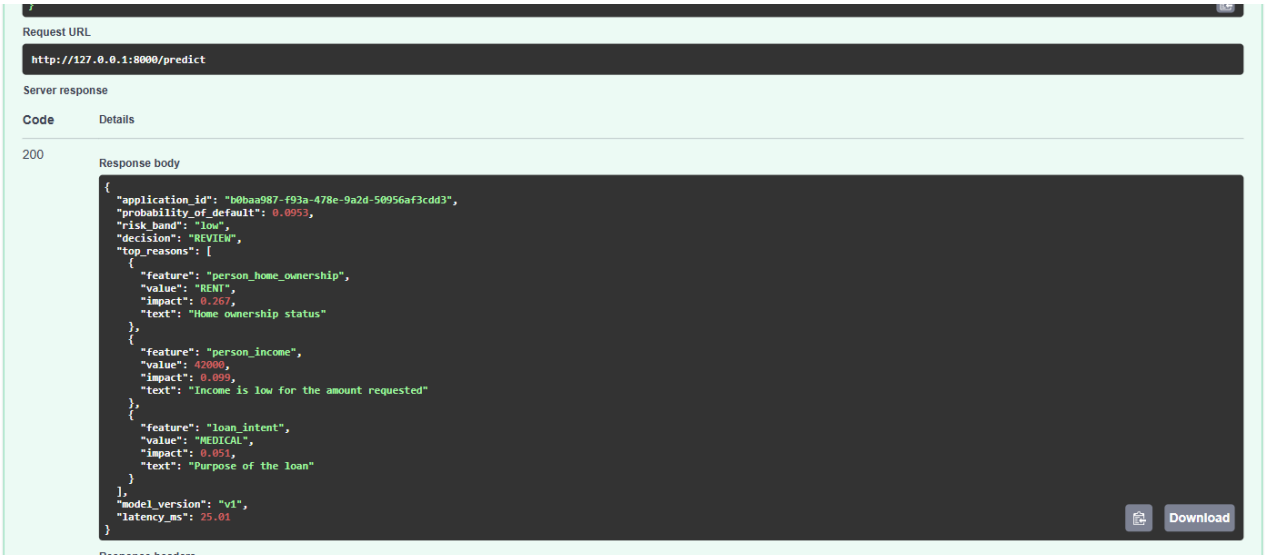


Screenshot 11: Evidently per-column drift table. Per-column detail of the same report: loan_int_rate (Wasserstein 0.597), loan_percent_income (0.410) and person_income (0.259) drifted; the other columns did not.

Kill switch

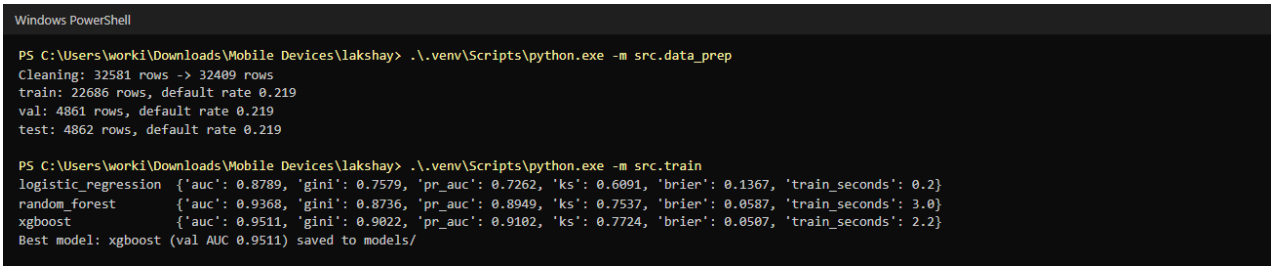


Screenshot 12: Kill switch on. The API started with SCORING_DISABLED=1 reports kill_switch true on GET /health.

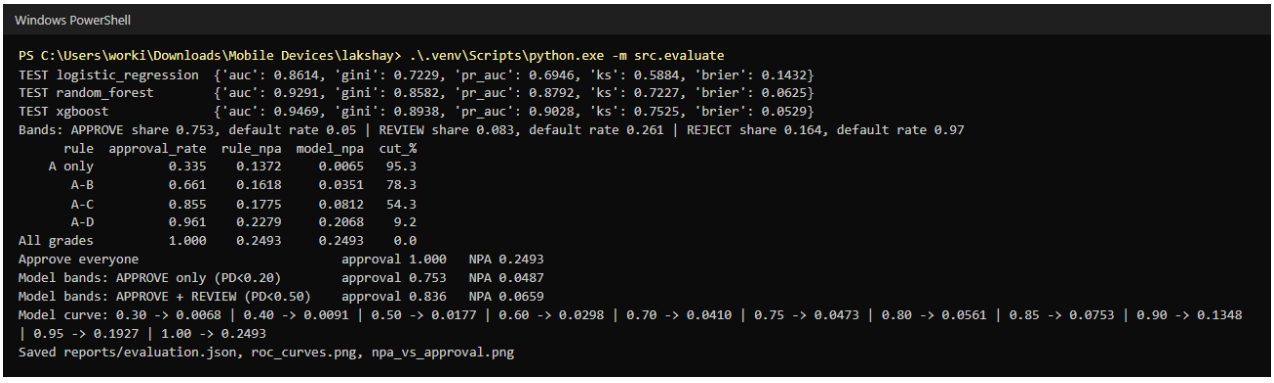


Screenshot 13: Every decision sent to a person. With the kill switch on, the low-risk applicant is still scored (PD 0.0953, band low) and logged, but the decision becomes REVIEW.

Pipeline runs (console)



Screenshot 14: Cleaning, splitting and training. Console output of src.data_prep (32,581 to 32,409 rows; train 22,686 / val 4,861 / test 4,862, default rate 0.219 each) and src.train (validation AUC: logistic regression 0.8789, random forest 0.9368, XGBoost 0.9511, which is saved as model_v1).



Screenshot 15: Test-set evaluation and NPA back-test. src.evaluate on the untouched test set: XGBoost AUC 0.9469; at the grade A-C rule's 85.5% approval rate the NPA ratio falls from 17.75% to 8.12% (54.3% lower).

```
Windows PowerShell

PS C:\Users\worki\Downloads\Mobile Devices\lakshay> .\.venv\Scripts\python.exe -m src.robustness
Cleaning: 32581 rows -> 32409 rows
logistic_regression  5-fold AUC mean 0.8707  sd 0.0059  min 0.8597
random_forest        5-fold AUC mean 0.9336  sd 0.0021  min 0.9310
xgboost              5-fold AUC mean 0.9497  sd 0.0024  min 0.9456
XGBoost without loan_grade and loan_int_rate: test AUC 0.9020
Saved reports/robustness.json
```

Screenshot 16: 5-fold cross-validation and ablation. src.robustness: XGBoost 5-fold AUC 0.9497 with standard deviation 0.0024; without loan_grade and loan_int_rate the test AUC is still 0.9020.

```
Windows PowerShell

PS C:\Users\worki\Downloads\Mobile Devices\lakshay> .\.venv\Scripts\python.exe -m src.explain
Mean |SHAP| (log-odds):
person_income                0.923
loan_grade                   0.892
loan_percent_income          0.827
person_home_ownership        0.687
loan_intent                  0.558
loan_int_rate                0.284
loan_amnt                    0.251
person_emp_length            0.170
person_age                   0.123
cb_person_cred_hist_length   0.052
cb_person_default_on_file    0.013
Base value -1.289 | max additivity error 1.34e-05
Riskiest test applicant: [{"feature": "loan_percent_income", "value": 0.56, "impact": 4.712, "text": "Loan is large compared with income"}, {"feature":
"person_income", "value": 14400, "impact": 4.309, "text": "Income is low for the amount requested"}, {"feature": "loan_intent", "value": "HOMEIMPROVEMENT",
"impact": 2.137, "text": "Purpose of the loan"}]
```

Screenshot 17: SHAP importance and reasons. src.explain: income, grade and loan-to-income ratio have the largest mean |SHAP|; base value -1.289 and maximum additivity error 1.34e-05; the riskiest test applicant gets three plain-language reasons.

```
Windows PowerShell

PS C:\Users\worki\Downloads\Mobile Devices\lakshay> .\.venv\Scripts\python.exe -m src.drift
                feature  psi_normal status_normal  psi_downturn status_downturn
             person_age      0.0027        stable        0.0027          stable
          person_income      0.0019        stable        0.1875           watch
      person_emp_length      0.0038        stable        0.0038          stable
             loan_amnt      0.0026        stable        0.0026          stable
         loan_int_rate      0.0024        stable        1.1231           alert
    loan_percent_income      0.0020        stable        0.1114           watch
cb_person_cred_hist_length      0.0025        stable        0.0025          stable
     person_home_ownership      0.0009        stable        0.0009          stable
            loan_intent      0.0013        stable        0.0013          stable
             loan_grade      0.0011        stable        0.0011          stable
 cb_person_default_on_file      0.0008        stable        0.0008          stable
          model_score_PD      0.0040        stable        0.1672           watch
normal: mean PD 0.218, approve 0.753, review 0.083, reject 0.164
downturn: mean PD 0.313, approve 0.612, review 0.145, reject 0.243
Saved reports/drift_summary.csv and reports/drift_report_downturn.html
```

Screenshot 18: Offline PSI table. src.drift compares the test set and its stressed copy with the training data: loan_int_rate PSI 1.1231 (alert) in the downturn, mean PD rising from 0.218 to 0.313.

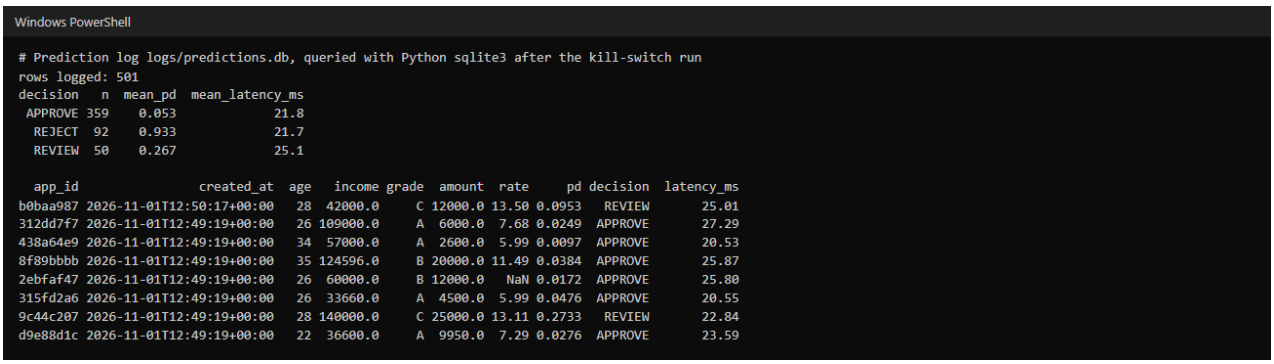
Traffic simulation, prediction log and tests

```
Windows PowerShell

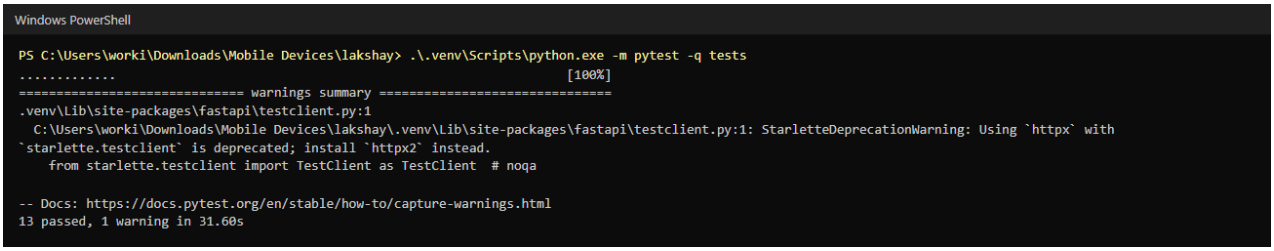
PS C:\Users\worki\Downloads\Mobile Devices\lakshay> .\.venv\Scripts\python.exe -m scripts.simulate_traffic --n 500
sent 500 requests | round-trip ms p50 35.9  p95 42.7
{'APPROVE': 0.718, 'REJECT': 0.184, 'REVIEW': 0.098}

PS C:\Users\worki\Downloads\Mobile Devices\lakshay> .\.venv\Scripts\python.exe -m scripts.simulate_traffic --n 500 --downturn
sent 500 requests | round-trip ms p50 34.6  p95 39.4
{'APPROVE': 0.572, 'REJECT': 0.274, 'REVIEW': 0.154}
```

Screenshot 19: Simulated loan-app traffic. scripts.simulate_traffic sent 500 normal and 500 downturn applications to the running API; in this run, made while recording the screenshots, round-trip p95 was 42.7 ms and 39.4 ms (the Phase 8 check run gave 45.6 ms and 44.9 ms), all under the 100 ms target.



Screenshot 20: SQLite prediction log. logs/predictions.db holds one row per scored application (501 rows here: 500 normal applications and the kill-switch request), with inputs, PD, decision and latency.



Screenshot 21: Automated tests. pytest runs the 13 tests (data, model and API) against a temporary database: 13 passed.